COLLEGE OF ENGINEERING

COLLEGE OF ENGINEERING (CE)

College Officers

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History

The College of Engineering at Rafik Hariri University (RHU) was established in 1999. Since then, the College has grown significantly and presently offers six undergraduate engineering programs in Biomedical, Civil, Computer and Communications, Electrical, Mechanical, and Mechatronics Engineering. Fully accredited by the Lebanese Ministry of Higher Education, the College works with each department and program towards accreditation by the Engineering Accreditation Commission of ABET – the most prestigious accrediting agency worldwide.

The College also features active research programs in several engineering areas. In this regard, master's degree programs are available in all six programs. For those students who are employed and can only be present on campus in the afternoons, a non-thesis master's degree option is also available. More than 400 students are currently enrolled in undergraduate and graduate programs.

Overview

The College of Engineering at RHU was established to address a perceived gap in high-caliber yet affordable higher education in technology and related fields. The College strives to apply learning methodologies that can transform entrant students into 21st-century leaders in their respective fields. Furthermore, the college stresses the importance of balancing knowledge, skills, and application. To that end, specialized Labs, modern classrooms, and a unique Co-op experience are implemented.

Vision

The College of Engineering aspires to be a center of excellence in engineering education and an incubator for innovation that serves Lebanon, the region, and the world.

Mission

The College of Engineering's mission is to provide quality professional education in a learning environment conducive to transformation, enabling our students to effectively innovate practical solutions to society's problems and individual needs in Lebanon, the Middle East, and the world.

Goals

We strive to achieve our mission. To this end, we are committed to continually revising our programs while keeping our senses attuned to our surroundings. More than anything else, our students - their learning and potential success - receive our greatest attention.

Values

The College of Engineering fully endorses the values of RHU of Academic Freedom of Inquiry, Excellence through Innovation, Ethical and Socially Responsible conduct, Tolerance and Diversity, Personal Initiatives and Individual Responsibility, Teamwork, Collaboration and cooperation, and Continuous Improvement.

Accreditation

The Bachelor of Engineering programs in Biomedical Engineering, Civil Engineering, Computer and Communications Engineering, Electrical Engineering, Mechanical Engineering, and Mechatronics Engineering have been accredited by the Engineering Accreditation Commission (EAC) of ABET, 415 North Charles Street, Baltimore, MD 21201; www.abet.org.

ABET is the most prestigious and internationally renowned engineering accreditation organization in the USA. ABET accreditation verifies that our educational experience meets the global standard for technical education in the engineering profession and enhances employment opportunities—multinational engineering corporations require graduation from an ABET-accredited program.

Academic Programs

The College of Engineering offers six undergraduate programs leading to a Bachelor of Science (BS) or a Bachelor of Engineering (BE) degree and six programs leading to a Master of Science (MS) degree.

Bachelor of Science Programs

1. Bachelor of Science in Civil Engineering
2. Bachelor of Science in Electrical Engineering
3. Bachelor of Science in Computer and Communications Engineering
4. Bachelor of Science in Biomedical Engineering
5. Bachelor of Science in Mechanical Engineering
6. Bachelor of Science in Mechatronics Engineering

Bachelor of Engineering Programs

1. Bachelor of Engineering in Civil Engineering
2. Bachelor of Engineering in Electrical Engineering
3. Bachelor of Engineering in Computer and Communications Engineering
4. Bachelor of Engineering in Biomedical Engineering
5. Bachelor of Engineering in Mechanical Engineering
6. Bachelor of Engineering in Mechatronics Engineering

Master of Science Programs

1. Master of Science in Civil Engineering
2. Master of Science in Electrical Engineering
3. Master of Science in Computer and Communications Engineering
4. Master of Science in Biomedical Engineering
5. Master of Science in Mechanical Engineering
6. Master of Science in Mechatronics Engineering

Program Codes

The following table lists the code used for each program. This code is used as the prefix of each course designation in the program.

Program Code	Program
CIVE	Civil Engineering
BIOM	Biomedical Engineering
CCEE	Computer and Communications Engineering
ELEC	Electrical and Electronics Engineering
MECH	Mechanical Engineering
MECA	Mechatronics Engineering

Admission Requirements

In addition to fulfilling the University admission requirements, students aspiring to study one of the engineering majors who did not submit SAT scores may be required to take placement examinations in mathematics and physics, depending on their Lebanese Baccalaureate scores in those subjects. Students who fail to attain a passing score in one of these subjects are required to take and pass a related remedial course to ensure their mastery of basic skills and improve their ability to handle the rigor of college-level subjects. Students are strongly advised to carefully review the University Catalog for admission and degree requirements, as well as all related academic policies.

Graduation Requirements

Each Engineering program offers students two undergraduate degrees: a Bachelor of Science (BS) or a Bachelor of Engineering (BE). The BS program encompasses 116 credits, and the BE 152 credits. A student must choose the degree that s/he wishes to attain before completing 60 credits. The first three years are common in both programs (except for the BS project). A student is eligible to receive a BS degree after completing the 116 credits cited in the first three years of the program's study plan. In addition to completing the credit requirements, a student is eligible to earn the BS or BE degree if s/he attains a Cumulative GPA, a Major GPA, and a Summative Learning Project Grade of 70 or above.

Students who completed and earned their BS degree from RHU and wish to pursue a BE degree in the same major may reapply for admission to the BE program. However, the student needs to take a technical elective course (3 credits) to replace the BS project.

The required credits are distributed among three categories: University requirements (UR), General Education (GE), College requirements (CR), and program requirements (PR). The University's General Education and College requirements are common to all departments in the College of Engineering. Each department has its own required, mandatory, and elective courses. The credit-hour allocations for each degree (BS or BE) within each program are shown in the following table.

Bachelor of Engineering (BE) degrees - 152 credits							
Program	General Education		College Requirements		Program Requirements		Total Credits
	Mandatory	Electives	Mandatory	Electives	Mandatory	Electives	
CIVE	20	10	27	0	80	15	152
BIOM	20	10	27	0	80	15	152
CCEE	17	13	27	0	77	18	152
ELEC	20	10	24	3	80	15	152
MECH	20	10	27	0	80	15	152
MECA	20	10	27	0	80	15	152
Credits	30		27		95		
Bachelor of Science (BS) degrees - 118 credits							

Program	General Education		College Requirements		Program Requirements		Total Credits
	Mandatory	Electives	Mandatory	Electives	Mandatory	Electives	
CIVE	20	7	24	0	67	0	118
BIOM	20	7	24	0	67	0	118
CCE	17	10	14	0	67	0	118
ELEC	20	7	21	3	67	0	118
MECH	20	7	24	0	67	0	118
MECA	20	7	24	0	67	0	118
Credits	27		23		67		118

An engineering student in all programs is eligible for graduation if s/he has:

1. Completed all the requirements of the degree
2. Attained a cumulative GPA of 70 or above.
3. Attained a major GPA of 70 or above.
4. Attained a grade of 70 or above on the Summative Learning Project.
5. Received a passing grade on the Co-op work experience.

I. General Education

Every BE (BS)- bound student is required to take 30 (27) credits of general education courses distributed across seven domains. 22 mandatory credit hours are taken from Domains 1, 4, 5, and 6 in addition to 9 (6 for B-bound students) elective credit hours from Domains 2 and 3, as indicated below.

Domain		Credits	Courses
1	Communication Competency*	9	ENGL 210: English Composition and Rhetoric ENGL 217: Professional English Communication ARAB 212: Arabic Language and Communication 1 credit- Community and Sustainability Elective
2	Humanities/Fine Arts	3	Selected from an approved list
3	Social Sciences/Culture	6	Selected from an approved list
4	Natural Sciences and Technology	6	3 Credits – All programs CCEE 214 or COSC 214 – Introduction to Programming 3 Credits Science Course: CHEM211-Environ.Chemistry-for CIVE+MECH+MECA and PHYS 312 – Modern Physics for ELEC+BIOM Science Elective for CCEE

5	Quantitative Reasoning	6	MATH 351 - Probability and Statistics ENGR 300 – Engineering Economics and Management

A description of the specific courses in this group is given below.

ARAB 212	**Arabic Language and Communication**	**2(2,0)**
This course helps students develop their ability to communicate effectively in standard Arabic. It provides students with the necessary Arabic communication skills they may need in their future jobs. Specifically, students learn how to write and orally present different forms of workplace documents in Arabic.		

CCEE 214	**Introduction to Programming + Lab**	**3(2,1)**
Understanding of the basic principles of programming and computer systems; Applications of programming to the solution of engineering problems; Control structures, functions, arrays, pointers, and structures. Laboratory experiments in programming for concepts. Equivalent to COSC 214.		

CMNS 200	**Etiquette**	**1(1,0)**
This course is designed to help students develop interpersonal and communication skills that are fundamental to success in the workplace, regardless of the industry, organization, or sector in which they are employed. Students will improve their professional style as they study topics including polite conversation, personal appearance, office politics, diplomacy, telephone, cell phone, and voicemail etiquette, meeting protocol, job interview presentation, and even international travel. Students will participate in an off-campus formal dining experience. **Co-requisite:** ENGL 210.		

CMNS 205	**First Aid**	**1(1,0)**
This course provides an overview of the principles of first aid, including preserving life, preventing further injury, and protecting the unconscious. It enhances students' knowledge of first-aid procedures and their application in real-life situations. The course also helps them develop the skills necessary to assist until medical help arrives. It raises students' awareness of different types of medical emergencies and the appropriate steps to take in both mild and life-threatening conditions. **Co-requisite:** ENGL 210.		

ENGL 210	**English Composition and Rhetoric**	**3(3,0)**
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, a research paper, and an oral presentation. **Prerequisite:** ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	**Professional English Communication**	**3(3,0)**
This required course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to		

write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.

ENGR 300	Engineering Economics and Management	3(3,0)
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics; present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting, and financial statements.		

MATH 351	Probability and Statistics	3(3,0)
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		

II. College Requirements

A. Remedial Programs and Courses

Proficiency in the English Language is a requirement for admission to any program in the College of Engineering. In addition, all incoming engineering students must sit for the placement exams in mathematics and physics. Depending on placement exam scores, a student may be required to take one or two remedial courses in Physics and/or Mathematics before taking program-level calculus and physics courses. The possible remedial courses are:

- MATH 190 - Calculus I
- MATH 191 – Calculus II
- PHYS 190 - General Physics – Mechanics (for MECH/MECA/CIVE students)
- PHYS 191 - General Physics - Electricity and Magnetism (for ELEC/CCE/BIOM students)

These remedial courses do not count toward fulfilling the degree requirements. A description of the remedial courses follows.

MATH 190	Calculus I	3(3,0)
Functions and their graphs, limits and continuity, differentiation, applications of derivatives, Extreme values and mean value theorem, definite and indefinite integrals, and the substitution method.		

MATH 191	Calculus II	3(3,0)
Covers limits and indeterminate forms with L'Hôpital's Rule, optimization, antiderivatives, and the Fundamental Theorem of Calculus. Topics include		

substitution methods, areas, volumes (cross-sections, shells), arc length, and surfaces of revolution. It introduces parametric and vector functions with applications to motion, arc length in space, curvature, and normal vectors. **Prerequisite**: Math 190.

PHYS 190	**General Physics – Mechanics**	**3(3,0)**
This course introduces students to the basic concepts and principles of Mechanics: Vectors, displacement, velocity, and acceleration; motion in 1D, projectile motion, forces, and Newton's laws of motion; work, energy, momentum, collision, torque, and rotational motion.		

PHYS 191	**General Physics - Electricity and Magnetism**	**3(3,0)**
This course is an introduction to electricity and magnetism. In this course, students will explore electric charges, electric forces and electric fields, electrical energy and potential, capacitance, direct-current circuits, magnetic force, magnetic field, induced voltage and inductance, alternating-current circuits, and electromagnetic waves.		

B. Mandatory Courses

All Engineering students (BS and BE) are required to take 24 credit hours of college-required courses. The BE program includes an additional mandatory 3-credit course, Engineering Project Management. A list of these courses and their descriptions follows.

Course	Title	Credits	Notes
Mathematics (18 credits for MECH, MECA, and CCEE; 15 credits for others)			
MATH 210	Discrete Mathematics	3	CCE major only
MATH 215	Calculus III	3	All majors
MATH 216	Calculus IV	2	All majors
MATH 311	Linear Algebra and Applications	3	All majors
MATH 314	Ordinary Differential Equations	3	All majors
MATH 317	Partial Differential Equations	3	MECH & MECA majors only
MATH 421	Numerical Analysis	3	All majors
Sciences (6 credits for MECH, MECA, and CCEE; 9 credits for all others)			
BIOL 210	Human Anatomy and Physiology and Lab	3	BIOM major only
GEOL 221	Fundamentals of Geology	3	CIVE major only
PHYS 210	Fundamentals of Physics and Lab	3	MECH, MECA, and CIVE majors only
PHYS 211	Physics: Electricity and Magnetism and Lab	3	ELEC, CCE, and BIOM majors only
	Science Elective	3	ELEC major
Engineering (6 credits)			
ELEC 210	Electric Circuits	3	All majors
ENGR 510	Engineering Project Management	3	All majors (BE track only)